

Flight Price Prediction System

Project Documentation

1. Dataset Description

The Flight Price Prediction dataset contains historical flight information and ticket prices. The objective is to predict the price of a flight ticket based on various flight-related features.

Features Used

- Airline
- Source City
- Departure Time
- Stops
- Arrival Time
- Destination City
- Travel Class
- Duration
- Days Left Before Departure
- Price (Target Variable)

The dataset was loaded using the Pandas library and explored to understand its structure, data types, and statistical properties.

2. Data Cleaning and Transformation

Missing Values

The dataset was checked for missing values using Pandas functions. No missing values were found.

Duplicate Records

Duplicate records were identified and checked. No duplicate records were present in the dataset.

Feature Encoding

Since machine learning models require numerical input, categorical features were transformed into numerical representations.

- One-Hot Encoding was applied to:
 - Airline
 - Source City
 - Departure Time
 - Arrival Time
 - Destination City
- Label Encoding was applied to:
 - Stops
 - Travel Class

Feature Selection

The Flight column was removed from the modeling process because it did not contribute significantly to ticket price prediction and increased dimensionality unnecessarily.

3. Exploratory Data Analysis (EDA)

Exploratory Data Analysis was performed to understand the relationships between variables and identify factors affecting flight prices.

Key Insights

- Business class tickets are significantly more expensive than Economy class tickets.
- Ticket prices generally increase as the departure date approaches.
- Flight duration has a positive relationship with ticket prices.
- Airlines exhibit different pricing patterns.
- The number of stops affects flight prices.
- Source and destination cities contribute to price variations.

Visualizations Used

- Histograms
- Boxplots
- Countplots
- Correlation Heatmaps
- Distribution Plots

These visualizations helped identify trends, distributions, and relationships within the dataset.

4. Model Development

Data Splitting

The dataset was divided into training and testing sets using an 80:20 ratio.

Models Implemented

- Linear Regression
- K-Nearest Neighbors (KNN)
- Decision Tree Regressor
- Random Forest Regressor
- Gradient Boosting Regressor
- AdaBoost Regressor
- Extra Trees Regressor
- Bagging Regressor
- Support Vector Regressor (SVR)
- XGBoost Regressor

Model Evaluation

The models were evaluated using:

- R^2 Score
 - Mean Absolute Error (MAE)
 - Mean Squared Error (MSE)
 - Root Mean Squared Error (RMSE)
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5. Hyperparameter Tuning

RandomizedSearchCV was used to optimize the Random Forest model parameters. Multiple parameter combinations were evaluated using cross-validation to improve model performance.

Cross-validation was also performed to assess the model's generalization ability.

6. Final Model Selection

After comparing all models, the Bagging Regressor achieved the best overall performance and was selected as the final model for deployment.

Final Performance:

- R^2 Score: 0.9853
 - MAE: 1086.23
 - MSE: 7,833,238
 - RMSE: 2798.79
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7. Model Deployment

The trained machine learning model was deployed using Streamlit.

Application Features

- User-friendly interface
- Interactive input fields
- Real-time flight price prediction
- Automatic preprocessing of user inputs
- Instant display of predicted ticket prices

The application runs locally using `streamlit run app.py`

8. Code Documentation

The source code contains comments explaining each major step, including:

- Dataset loading
 - Data preprocessing
 - Feature encoding
 - Model training
 - Hyperparameter tuning
 - Model evaluation
 - Model saving and loading
 - Streamlit deployment
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9. Conclusion

This project successfully implemented an end-to-end machine learning solution for flight price prediction. The workflow included data preprocessing, exploratory data analysis, model development, hyperparameter tuning, evaluation, and deployment using Streamlit. The final application allows users to predict flight prices efficiently through an interactive web interface.